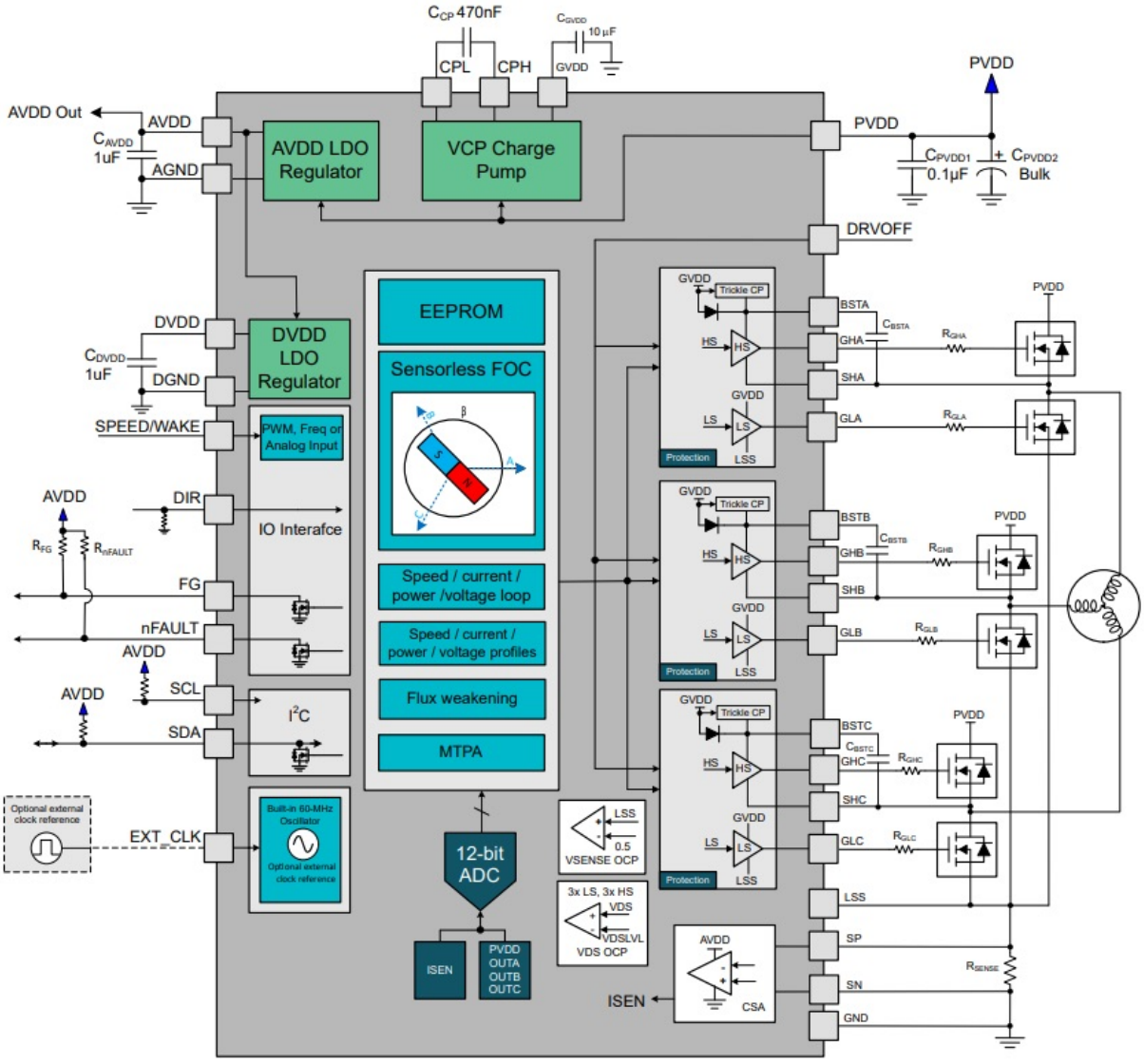


Figure 6-1. MCF8329A-Q1 Functional Block Diagram

Table 6-1. MCF8329A-Q1 External Components

COMPONENTS	PIN 1	PIN 2	RECOMMENDED
C_{PVDD1}	PVDD	GND	X7R, 0.1µF, >2x PVDD-rated
C_{PVDD2}	PVDD	GND	≥ 10µF, >2x PVDD-rated
C_{CP}	CPH	CPL	X7R, 470nF, PVDD-rated
C_{AVDD}	AVDD	AGND	X7R, 1µF or 2.2µF, 10V
C_{GVDD}	GVDD	GND	X7R, ≥10uF, 30V
C_{DVDD}	DVDD	DGND	X7R, 1µF, 10V
C_{BSTx}	BSTx	SHx	X7R, 1µF, 25V
R_{nFAULT}	1.8 to 5 V Supply	nFAULT	5.1kΩ, Pullup resistor
R_{FG}	1.8 to 5 V Supply	FG	5.1kΩ, Pullup resistor
R_{SDA}	1.8 to 5 V Supply	SDA	5.1kΩ, Pullup resistor
R_{SCL}	1.8 to 5 V Supply	SCL	5.1kΩ, Pullup resistor
R_{DIR}	DIR	AGND	Optional ≤ 10kΩ resistor for better noise immunity, if DIR pin is used

R_{sens} :

$$R_{sens} = \frac{V_{SO} - \frac{V_{REF}}{2}}{CSA_GAIN \cdot 2}$$

$V_{Ref} = 3\,V$
$0{,}3\,V < V_{SO} \leq 3\,V$
$I \rightarrow$ max piekstroom v/d motor

Voor hoge stromen $\rightarrow$ CSA_gain $\rightarrow$ Laag
Zodat V_{SO} niet meteen maar 3 V springt

Berekening :

$$\frac{V_{SO} - \frac{V_{Ref}}{2}}{C_{SA}\text{-gain} \cdot I} \quad | \rightarrow \begin{cases} V_{SO\,max} = 2.8\,V & \text{(Veiligheids Marge)} \\ V_{Ref} = 3\,V \\ C_{SA}\text{-gain} = 20 \end{cases}$$

CSA_gain = 20 $\rightarrow$ $V_{shunt} \cdot 20$

$V_{shunt} = 100\,A \cdot 0{,}0005 = 0{,}05\,V$
$V_{out} = (I \cdot R_{sens}) \cdot 20 = 1\,V$

⊕ Ruis word ook versterkt - maar niet 40 keer
⊕ Stabieler bij schakelen
⊖ Meer vermogen verlies dan gain 40
to minder Ruisgevoelig

$R_{sens} = 0,325\ m\Omega$

$P = I^2 R = 100 \cdot 0,000325 = 3,25\ W$

$\rightarrow$ R_{sens} $\rightarrow$ Shunt Resistor $0,3\ m\Omega$ $5\ W$

RGHx
RGLx
→ • Beperken piek-gate-stroom
• Regelen Schakelsnelheid
• Verminderen Ringing / EMI
• Beschermen Tegen te hoge Pick stromen

Berekenen:

1. Gate-drive limiet van het IC

2. Basis Formule:

$$I_{gate} = \frac{V_{DRV}}{RG + R_{int}}$$

dus: $$RG = \frac{V_{DRV}}{I_{Gate}} - R_{int}$$

V_{DRV} → 10-12 V (gVDD / bus gate drive)

Igate = gewenste laadstroom

Rint = interne Driver weerstand

1.

$Q_G = 26\ nC$

t_{sw} = gewenste schakeltijd (50-200 ns)

$$I_{gate} = \frac{Q_G}{t_{sw}}$$

2. GATE weerstand

$$R_G = \frac{V_{DRV}}{I_{gate}} - R_{driver}$$

- V_{DRV} (10-12 V)
- R_{driver} (1-3 Ω)

3. zonder combinatie

$$R_G = \frac{V_{DRV} \cdot t_{sw}}{Q_g}$$

$$R_G = \frac{12\ V \cdot 100\ ns}{26\ nC} =$$

$R_{GH} \approx 22 - 33\ \Omega$

$R_{GL} \approx 10 - 15\ \Omega$